

Alan Hsu

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Junior at Purdue University passionate about making humans spacefaring, looking for ownership of dirt cheap high reliability avionics.

Education

Purdue University, School of Electrical and Computer Engineering, West Lafayette, IN *Dec 2027 | 3.95 / 4 GPA*

- Bachelor's of Science in Electrical Engineering

Thomas Jefferson High School for Science and Technology, Alexandria, VA *June 2023 | 4.450 / 4 GPA*

Work Experience

Starpath Robotics | Web: starpath.space

Avionics Engineer (Part Time) | Offsite; West Lafayette, IN *Aug 2025 – Present*

- Continuing to develop and qualify motor control and battery management systems
- Advising on thermal and radiation trades to be made on Rover 11, Starpath's first flight rover
- Trading high level decisions for power management for 100kW vertical solar array and processing plant

Interim Avionics Lead (Co-op position) | Hawthorne, CA *May 2024 – Aug 2025*

- Set scope, timelines, and budgets and kicked off Starpath's Avionics program, which will bring avionics systems from terrestrial engineering pathfinder to lunar flight readiness and scalability by 2027
- Collaborated with Rover Mechanical team to close the thermal model and negotiate relevant thermal requirements for a first of its kind lunar PSR capable and lunar night surviving mining rover
- Designed and built all avionics systems, including for Starpath's submission to NASA BTIL Level 3, Starpath's first full system integrated demo, and Starpath's first dusty TVAC test of Rover in MSFC's V20 25ft chamber
- Pushed over 70 total PCB revisions across all projects since May '24. Notable projects include:
 - Radiation tolerant GaN based 60V 5kWpk capable FOC BLDC motor controllers
 - Radiation tolerant GaN based power management and distribution and battery management for a 15kWh 60V Li-Ion pack
 - Embedded architecture and software implementation for Chassis TVAC Test Article, Rover 10, and Plant 2, each involving CAN communications between STM32 processors in C
 - Custom lunar capable LiDAR iToF imager PCB, in collaboration with engineers from NASA GSFC
 - Link budgeting and system design for a long range S-band mesh network for a fleet of rovers
- Managed one intern in summer of 2024 responsible for hardware for Starpath's first proton beam tests

Electrical Engineering Intern | San Francisco, CA *June – Aug. 2022 | June – Aug. 2023*

- Integrated and harnessed electronics for Rovers 3, 4, and 8, which competed in various phases of NASA BTIL

Extracurriculars

Purdue FLaC-Sat VIP | FEMTA-Sat Team *Jan. 2024 – May 2024*

- Owned test, debug, and iteration on control circuitry for UPW thrusters on the FEMTA-Sat Cubesat
- Developed microcontroller based control system for thrust control and thermal management of UPW propellant

Purdue Solar Racing | Solar Subteam *Aug. 2023 – May 2024*

- Prototyped a low cost high-voltage MPPT boost converter for Lux, PSR's submission to ASC 2024

TJ UAV Club | Competition Team Captain & Electronics Lead *Sept. 2021 – June 2023*

- Led electrical development for Avalon X, a fixed wing aircraft that competed in the 2023 SUAS competition; led flight line operations, safety, and logistics at competition
- Designed and built two iterations of the aircraft's electronics bay and a self-stabilizing camera gimbal

TJ Space Program | TJREVERB Technical Lead *Sept. 2021 – June 2023*

- Led electrical development for TJREVERB, a 2U CubeSat deployed on December 29, 2022 to determine the feasibility of Iridium Short Burst Data (SBD) as a telemetry radio for CubeSats
- Assisted in final assembly, vibration testing, and final integration into Nanoracks deployer
- Designed, built, and iterated a Raspberry Pi and Iridium-based CubeSat bus to provide a low cost and easy to use platform for future missions; coached underclassmen in mission and hardware design

Skills

- Altium Designer, KiCAD, Fusion 360, Solidworks, LTSpice, Ubuntu, Python, C/C++, MATLAB
- Design for survival in High Vibration, Radiation, and Thermals; Data, Link, and Power Budgeting